Moral hazard—Principal agent model

Principal-Agent. Introduction

- The principal ``controls” an operation in which the agent participates.
- The agent’s actions have consequences for the principal.
- The principal does not observe perfectly the agent’s actions.
- The principal designs an incentive system to best align the agent’s actions with the principal’s objectives.

- Example: An owner (principal) hires an agent to manage a restaurant; the manager may work hard or not; the owner cannot perfectly monitor the manager’s effort.
- Moral hazard: Asymmetry of information arises from a ``hidden” action—an agent’s action that is not perfectly observable by the principal.

Adverse selection: Asymmetry of information arises from information only available to one party—private information.

Applications

- See Bolton and Dewatripont (2005).
- Survey, The theory of contracts, Hart and Holmström (1987).
- Managerial theory of the firm; Jensent and Meckling (1976), Grossman and Hart (1983).
- Optimal contracts between landlords and tenants, sharecroping.
- Efficiency wage theories.
- Financial intermediation; Diamond (1984).
- Theoretical Accounting; Demski and Kreps (1982).

Models

There are at least three representations of the agency model (Hart and Holmström (1987)).

- The *state space* formulation,
- The *parametrized* formulation,
- The *abstract* formulation.

State space formulation, a worker

- The owner of a firm, the principal, hires a worker or manager, the agent.
- The firm's technology produces a publicly observable output y.
- $y = y(\theta, e)$ where $y : \Theta \times A \to \mathbb{R}$.
  For simplicity $y = \theta + e$.
  - $\theta \in \mathbb{R}$, unobservable random variable with density $f(\theta) > 0$
  - $e \in A \subset \mathbb{R}$ agent's effort level, unobservable to principal.

- $t : \mathbb{R} \to \mathbb{R}$, money transfer $t(y)$ from the principal to the agent when output is y; compensation package.
- Principal's utility: $U_p(y - t(y))$, $U_p' > 0, U_p'' \leq 0$.
- Agent's utility: $U_a(t) - \Psi(e)$, where $U_a' > 0$, $U_a'' \leq 0$ (concave), and $\Psi' > 0$ and $\Psi'' \geq 0$ (convex).
  $\Psi(e)$ is the cost to the agent of exerting effort e.

  It is common in the literature to assume that the agent's utility is separable in income and effort.
- $\bar{U}_a$ is the agent's reservation utility.

The principal-agent model—Timing

1. The principal proposes a contract $t(y)$ where y is observable output and t is the payment to the agent.
2. If the agent rejects the contract, the game ends. Players receive their reservation utility.
3. If agent accepts, agent chooses his action (effort level) $e \in A$. The choice e is not contractable.
4. Output $y(\theta, e)$ is realized and observed, where θ is an unobserved random variable with known distribution.
5. The agent receives $t(y)$, the principal receives $y - t(y)$ and the game ends.

Complete information. Benchmark

- The principal's problem is to find a function t (that depends only on the observable variable y and that maximizes the principal's utility inducing the agent to participate in the process (i.e. IR).
- For comparison, it is useful to consider the complete information case, i.e. the effort level $e \in A$ is observable and contractable.

Principal's problem, full information

$$
\begin{aligned}
&\max_{t,e} && E_\theta U_p(y - t(y)) \\
&s.t. && y = y(\theta + e) \\
&(IR) && E[U_a(t) - \Psi(e)] \geq \bar{U}_a \\
& && e \in A \\
& && t \in L^1(\mathbb{R})
\end{aligned}
$$

In an equivalent formulation the Principal chooses a transfer function $t(y, e)$.

For definition of L_1, see for instance Royden (1968).

First order conditions

$$\mathcal{L}(t, e, \lambda) = \int U_p(y - t(y))\, f(\theta)\, \mathrm{d}\theta + \lambda \left[\int U_a(t(y))\, f(\theta)\, \mathrm{d}\theta - \Psi(e) - \bar{U}_a \right]$$

First order conditions with respect to t and e,

$$0 = -U_p' f + \lambda U_a' f \tag{1}$$

$$0 = \int U_p' \times \underbrace{\left(\frac{\partial y(\theta, e)}{\partial e} \right)}_{\frac{\partial(\theta + e)}{\partial e} = 1} f\, \mathrm{d}\theta - \lambda \Psi'(e) \tag{2}$$

FOC with respect to t and e,

$$\begin{aligned} 0 =& -U_p' f + \lambda U_a' f \\ U_p' =& \lambda U_a' \end{aligned}$$

where f is eliminated because $f > 0$.

$$0 = \int U_p' f \, \mathrm{d}\theta - \lambda \Psi'(e)$$

Aside: order of derivation and integration

Let $f : X \times A \to \mathbb{R}$ such that $\forall x \in X$, $f(x, \cdot)$ is differentiable with respect to a; and $\forall a \in A$, $f(\cdot, a)$ is integrable. For a distribution μ on X, define $G(a) = \int f(x,a)\,\mathrm{d}\mu$, then $\frac{\mathrm{d}G}{\mathrm{d}a} = \lim_{h\to 0} \frac{G(a+h)-G(a)}{|h|}$.

$$\begin{aligned}
\frac{\mathrm{d}G}{\mathrm{d}a} &= \lim_{h\to 0} \int \frac{f(x,a+h) - f(x,a)}{|h|}\,\mathrm{d}\mu \\
&=^? \int \lim_{h\to 0} \frac{f(x,a+h) - f(x,a)}{|h|}\,\mathrm{d}\mu \\
&= \int \frac{\mathrm{d}f(x,a)}{\mathrm{d}a}\,\mathrm{d}\mu
\end{aligned}$$

- Let $g^h(x,a) = \frac{f(x,a+h)-f(x,a)}{|h|}$.
- Since f is differentiable wrt a, for fix a $\lim_{h\to 0} g^h(x,a) = \frac{\mathrm{d}f(x,a)}{\mathrm{d}a}$ pointwise.
- The Dominated Convergence Theorem completes the argument.

Aside: Gateaux directional derivative

Definition 1 (Gateaux differential) Let X be a vector space, Y a normed space, and $T : X \to Y$. The Gateaux differential of T at $x \in X$ with increment $h \in X$ is

$$\gamma T(x; h) = \lim_{\alpha \downarrow 0} \frac{1}{\alpha}[T(x + \alpha h) - T(x)]$$

if it exists.

If the limit exists for every $h \in X$, the function T is said to be Gateaux differentiable at x.

RN principal, RA agent, observable e

Risk-neutral principal, risk-averse agent, observable e.

- Wlog let $U_p(y - t) = y - t(y)$ and $U_p' = 1$.

Then Equation 1, FOC w.r.t t, becomes

$$\begin{aligned} -U_p' + \lambda U_a' &= 0 \\ \lambda &= \frac{U_p'}{U_a'} \\ U_a'(t(y)) &= \frac{1}{\lambda}\ \forall y \end{aligned}$$

- Then optimal contract $t(y) = k$, a constant.

FOC w.r.t. e used to determine the optimal e^*.

$$0 = \int U_p' \times \left(\frac{\partial y(\theta, e)}{\partial e} \right) f \, \mathrm{d}\theta - \lambda \Psi'(e)$$

$$0 = 1 - \lambda \Psi'(e) \text{ because } U_p' = 1$$

$$\Psi'(e) = \frac{1}{\lambda}$$

Since $\frac{1}{\lambda} = U_a'(t(y))$, the optimal effort level e^* is the solution to

$$\Psi'(e^*) = U_a'(t(y)) = U_a'(k)$$

Inefficiency of $t(y) = k$ with unobservable e

$$\max_{e \in A} EU_a(t) - \Psi(e)$$

$$\max_{e \in A} EU_a(k) - \Psi(e)$$

The FOC wrt to ``e” is

$$-\Psi'(e) = 0$$

QED

- The optimal contract under imperfect information was not derived.
- Thus, previous slide does not ``prove” that the first best cannot be achieved under imperfect information.
- That conclusion follows, however, because the only way to provide complete insurance to the risk-averse agent is with a constant $t(y) = k$.
- Constant transfer would necessarily elicit lower effort level than it is optimal under complete information.

RA principal, RN agent, observable e

- W.l.o.g. let $U_a(t(y)) = t(y)$ and $U_a' = 1$.

From FOC Equation 1, since $f > 0$, it follows that

$$\forall y,\ U_p'(y - t(y)) = \lambda$$

This implies that $y - t(y) = K$, a constant.

- From FOC Equation 2, $1 - \Psi'(e^*) = 0$.
- Single instrument and single objective, risk sharing. With incomplete information there are two objectives, risk sharing and correct incentives to work.

Theorem 1 Suppose the effort level $e \in A$ is not observable by the principal, the principal is risk averse (RA-P) and the agent is risk neutral (RN-A). Then, there exists a contract that achieves the first best solution, i.e., the solution of the complete information case.

- Intuition: Since the agent is risk neutral, risk sharing is not an issue. Thus $t(y)$ need only elicit the correct effort level from the agent.

RA-Principal, RN-A, unobservable e

- Let e^*, $t^*(y)$ be the optimal solution under complete information.
- Then, $y - t^*(y) = K$ and therefore $t^*(y) = y - K$.
- The value of K must be determined.

The value of K is obtained from IR.

$$
\begin{aligned}
EU_a(t(y)) - \Psi(e^*) &= \bar{U}_a \\
EU_a(t(y)) &= \Psi(e^*) + \bar{U}_a \\
Et(y) &= \Psi(e^*) + \bar{U}_a \\
Ey - K &= \Psi(e^*) + \bar{U}_a \\
e^* + E\theta - K &= \Psi(e^*) + \bar{U}_a \\
e^* + E\theta - \Psi(e^*) - \bar{U}_a &= K
\end{aligned}
$$

Therefore

$$
t^*(y) = y - K = y + \bar{U}_a + \Psi(e^*) - e^* - E\theta
$$

It remains to verify that with t^*, agent chooses e^*.

$$\max_{e \in A} EU_a(t) - \Psi(e)$$

$$\max_{e \in A} Et^*(y) - \Psi(e)$$

$$\max_{e \in A} Ey + \bar{U}_a + \Psi(e^*) - e^* - E\theta - \Psi(e)$$

$$\max_{e \in A} E\theta + e + \bar{U}_a + \Psi(e^*) - e^* - E\theta - \Psi(e)$$

The FOC wrt to ``e'' is, $1 - \Psi'(e) = 0$.

QED

Preliminaries.

- For the definitions and basic results on stochastic dominance see, for instance, Mas-Colell et al. (1995), page 195-197.
- First- and second-order stochastic dominance were introduced to economics in Rothschild and Stiglitz (1976).
- See also Marshall et al. (2010), Chapter 17, pages 707-712.

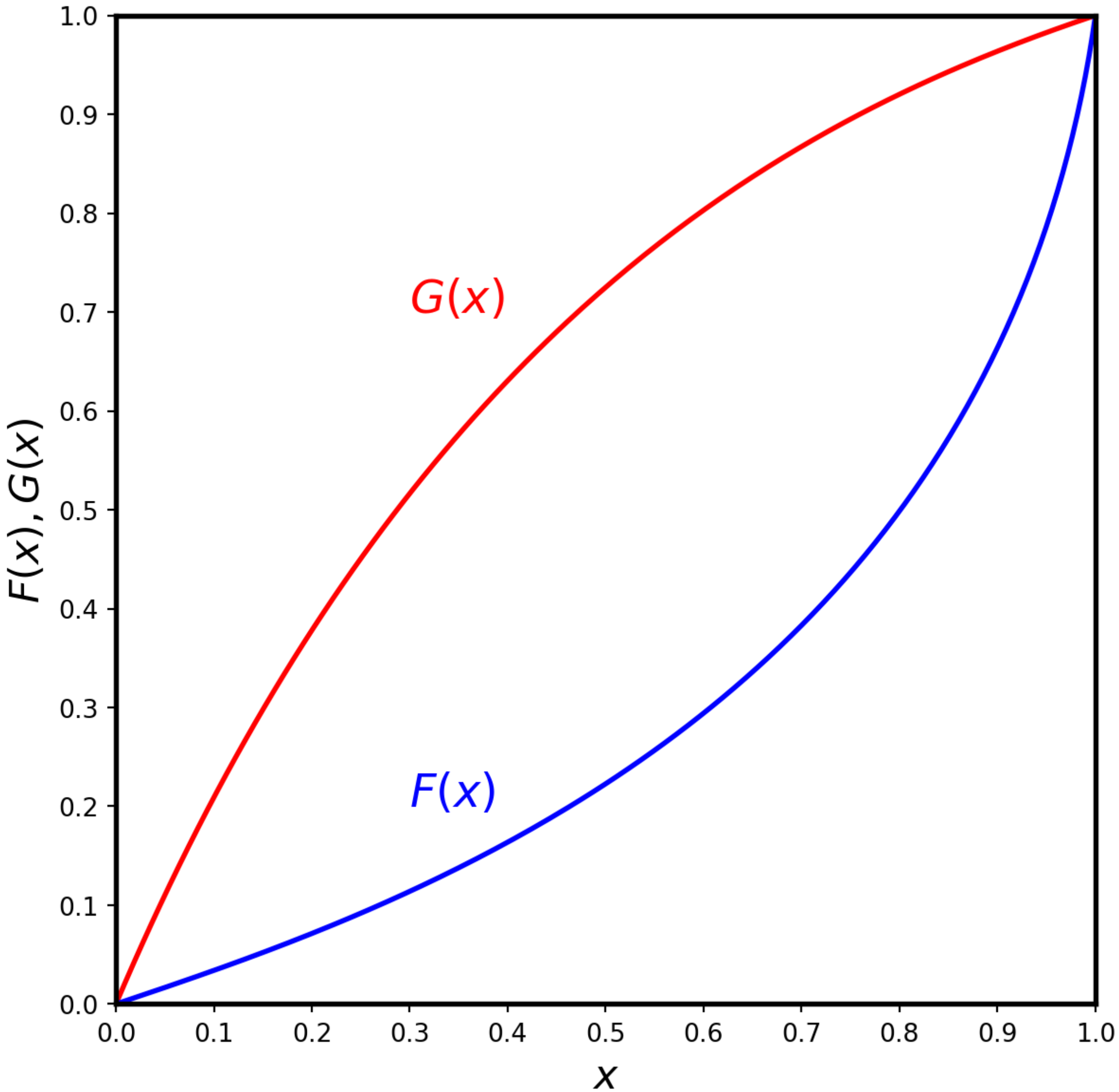
FOSD

Definition 2 (First-order stochastic dominance) Let μ, ν be two probability measures on $X \subset \mathbb{R}$.

We say that μ first-order stochastically dominates ν if for every nondecreasing function $u : \mathbb{R} \to \mathbb{R}$, $\int u(x)d\mu \geq \int u(x)d\nu$.

Theorem 2 The cummulative distribution function (cdf) F first-order stochastically dominates the cdf G if and only if $(\forall x)\ F(x) \leq G(x)$.

FOSD



$$F \succ_{FOSD} G$$

$$(\forall x), 1 - F(x) > 1 - G(x)$$

SOSD

Definition 3 (Second-Order Stochastic Dominance) Let F, G be two cdf with the same mean. F second-order stochastically dominates (or is less risky than) G if for every nondecreasing concave function $u : \mathbb{R}_+ \to \mathbb{R}$, $\int u(x)dF(x) \geq \int u(x)dG(x)$.

Definition 4 (Mean-Preserving Spread) Let F be the cdf corresponding to a random variable x. For any realization x, let $H_x(z)$ be the cdf corrsponding to a random variable z and suppose $\int zdH_x(z) = 0$. Consider the new random variable $y = x + z(x)$ and let G be its cdf. Then we say that G is a mean-preserving spread of F.

Theorem 3 Let F, G be two cdfs with the same mean. The following statements are equivalent.

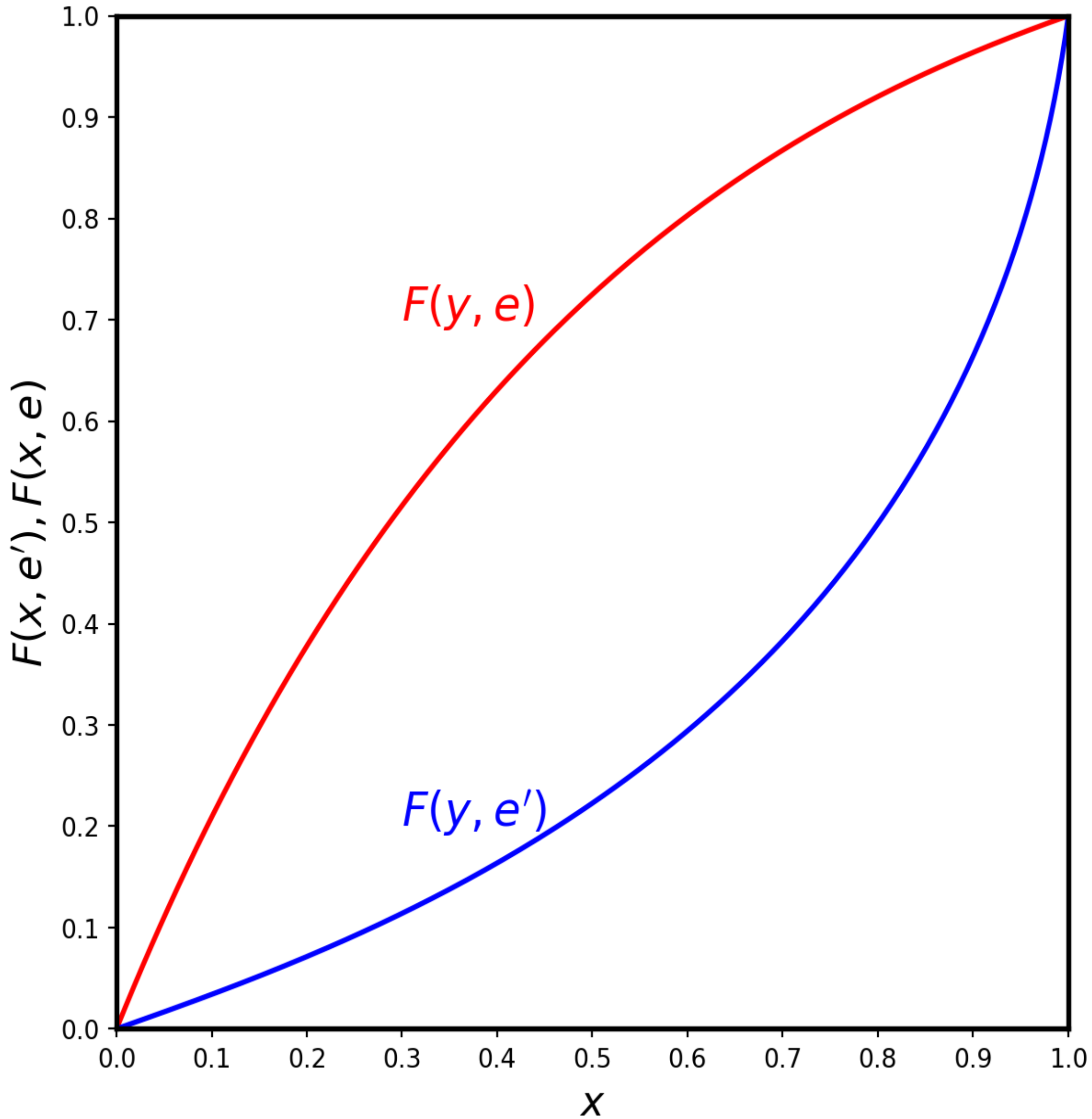
1. F second-order stochastically dominates G.
2. G is a mean-preserving spread of F.
3. $\int_0^x G(t)dt \geq \int_0^x F(t)dt, \forall x$.

Parametrized distribution approach

- It provides intuitive necessary conditions for the optimal mechanism.
- Output is a random variable $y \in [\underline{y}, \bar{y}]$ with twice continuously differentiable cdf $F(y, e)$ and density $f(y, e)$.
- First order stochastic dominance FOSD implies that
  $e' > e \implies F(x, e') \leq F(x, e)$.

  Higher values of e yield probabilistically higher y.
- $\frac{f(y,e')}{f(y,e)}$ must be bounded, see Mirrlees (1999).

FOSD in parametrized model



$(e' > e)$ implies

$$F(\cdot, e') \succ_{FOSD} F(\cdot, e)$$

$$(\forall y)\ 1 - F(y, e') > 1 - F(y, e)$$

Principal's problem

$$\max_{t(y),e} \int U_p(y - t(y)) f(y,e) dy$$

$$s.t.$$

$$(IR) \qquad \int U_a(t(y)) f(y,e) dy - \Psi(e) \geq \bar{U}_a$$

$$(IC) \qquad e \in \arg\max_{e' \in A} \left\{ \int U_a(t(y)) f(y,e') dy - \Psi(e') \right\}$$

The IC constraint can be replaced by the first and second order conditions of the implicit agent's maximization problem:

$$\int U_a(t(y))\frac{\partial f(y,e)}{\partial e}dy = \Psi'(e) \tag{3}$$

$$\int U_a(t(y))\frac{\partial^2 f(y,e)}{\partial e^2}dy \leq \Psi''(e) \tag{4}$$

First order approach

- Some authors ignore SOC (Equation 4) and use only FOC (Equation 3), for instance Holmström (1979), Shavell (1979).
- The first order approach is not legitimate; it may identify a minimum instead of a maximum, or the equivalent of an inflection point.
- We will first follow the first order approach, then illustrate why it may lead to incorrect results, and finally give conditions under which it is valid.

FOA. Program

Replacing the IC constraint in the seller's problem by the first-order condition of the agent's problem one obtains a modified program,

$$\max_{t(y),e} \int U_p(y - t(y)) f(y,e) dy$$

$$s.t.$$

$$\int U_a(t(y)) f(y,e) dy - \Psi(e) \geq \bar{U}_a$$

$$\int U_a(t(y)) \frac{\partial f(y,e)}{\partial e} dy = \Psi'(e).$$

FOA. Lagrangean

$$\mathcal{L}(t, e, \lambda, \mu) = \int \{U_p(y - t(y))f(y, e) + \lambda \left[U_a(t(y))f(y, e) - \Psi(e) - \bar{U}_a\right] + \mu \left[U_a(t(y))f_e(y, e) - \Psi'(e)\right]\} \, dy$$

where λ and μ are the multipliers of the IR and IC constraints respectively.

FOA. First order condition, $t(\cdot)$

$$
\begin{aligned}
-U_p'(y - t(y)) f(y,e) + \lambda U_a'(t(y)) f(y,e) \\
+ \mu U_a'(t(y)) f_e(y,e) = 0
\end{aligned}
$$

Dividing by $U_a'(t(y)) f(y,e)$ yields the necessary condition

$$
\frac{U_p'(y - t(y))}{U_a'(t(y))} = \lambda + \mu \frac{f_e(y,e)}{f(y,e)}. \tag{5}
$$

Observation 1

- Grossman and Hart (1983) propose a two-step method for solving PA problems in the parametrized distribution approach.

1. For each e, find $t(y)$ that minimizes expected $t(y)$ subject to agent's IR and IC.
2. Find e that maximizes expected output minus the expected cost.

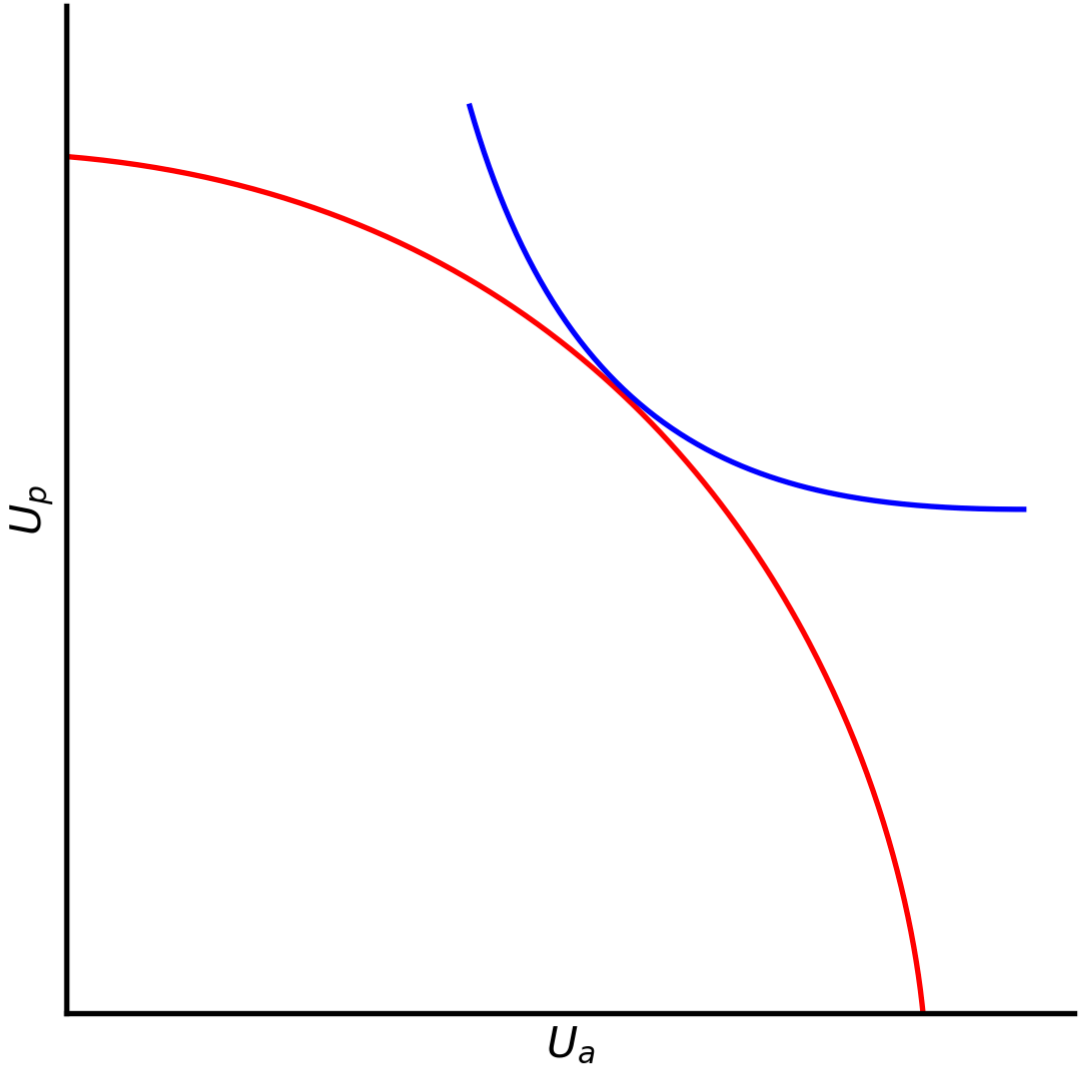
Observation 2

- If the ``surrogate” IC constraint is not binding, $\mu = 0$. Then Equation 5 becomes

$$\frac{U_p'(y - t(y))}{U_a'(t(y))} = \lambda$$

This is a necessary and sufficient condition for optimal risk sharing (Borch (1962)).

- Intuition. The multiplier λ is the utility cost to the principal of providing more utility to the agent.



$$\lambda = \frac{U'_p}{U'_a} = \frac{dU_p/dt}{dU_a/dt} = \frac{dU_p}{dU_a}$$

Observation 3

- Under the maintain assumptions, if $F_e(y, e) \leq 0$ with strict inequality for some values of e, then $\mu > 0$.
- Proof. Differentiate the Lagrangean with respect to e. Make use of second order condition on the Agent's problem.
- There is a tradeoff between optimal risk sharing and the provision of incentives for the agent.

Observation 4

- Interpretation of Equation 5 by analogy with two action case.
- There are only two effort levels $e_l < e_h$.
- If the principal wishes to implement $e = e_l$, the problem is simple: t must be such that IR is satisfied.
- If the principal wishes to implement $e = e_h$, in addition to IR, t must satisfy IC.

$$\int U_a(t(y))f(y,e_h)dy - \Psi(e_h)$$
$$\geq \int U_a(t(y))f(y,el)dy - \Psi(el)$$

$$\int U_a(t(y))[f(y,e_h) - f(y,el)]dy$$
$$- [\Psi(e_h) - \Psi(el)] \geq 0$$

- Replace IC by equation above and differentiate new Lagrangean w.r.t. t.

Assuming principal wants $e = e_h$, then

$$\mathcal{L} = \int \{U_p f(y, e_h) + \lambda \left[U_a f(y, e_h) - \Psi(e_h) - \bar{U}_a\right]$$
$$+ \mu \int U_a(t(y))[f(y, e_h) - f(y, e_l)]\} \, dy$$
$$- \mu[\Psi(e_h) - \Psi(e_l)]$$

FOC w.r.t. t

$$-U_p' f(y, e_h) + \lambda U_a' f(y, e_h) + \mu U_a' [f(y, e_h) - f(y, e_l)] = 0$$

from which one obtains the necessary condition

$$\frac{U_p'(y-t(y))}{U_a'(t(y))} = \lambda + \mu\left[1 - \frac{f(y,e_l)}{f(y,e_h)}\right]$$

If principal is risk neutral, $U_p' = 1$. Then

$$\frac{1}{U_a'(t(y))} = \lambda + \mu\left[1 - \frac{f(y,e_l)}{f(y,e_h)}\right]$$

- The equation ``says” that $t(y)$ varies with y.
- When $\frac{f(y,e_l)}{f(y,e_h)} < 1$ then e_h is more likely than e_l.

$$\frac{1}{U_a'(t(y))} = \lambda + \mu\left[1 - \frac{f(y, e_l)}{f(y, e_h)}\right]$$

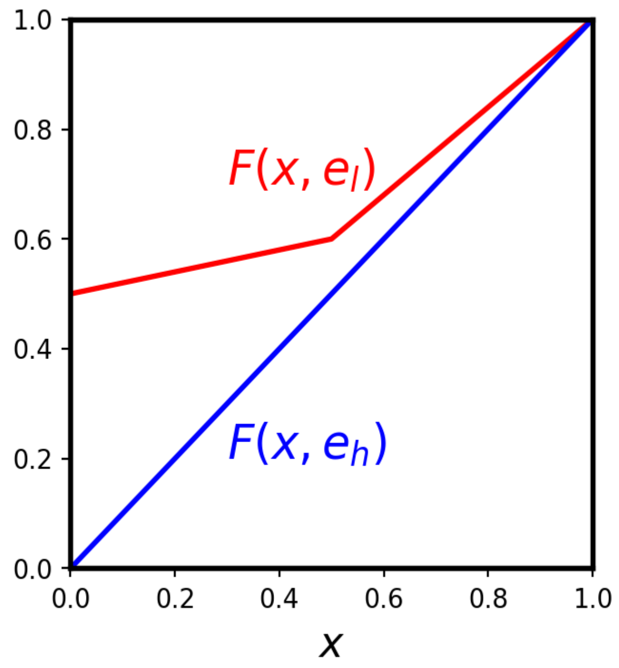
- Intuition. Suppose higher y {} that e_h is more likely than e_l, then $\frac{f(y,e_l)}{f(y,e_h)}$ becomes smaller.
- This means RHS is larger.
- In turn this must imply that U_a' is smaller.
- Smaller U_a' means larger $t(y)$ because U_a is concave.

Observation 5

- $\frac{f(y,e_l)}{f(y,e_h)}$ is the likelihood ratio; it indicates the likelihood that the agent chooses e_l relative to e_h.
- Agent is punished when outcomes move beliefs about e_h down and rewarded in opposite case.
- Agency problem is not an inference problem in the statistical sense. The principal gains no new information about the action taken by the agent from observing y because, by contract design, the principal already knows the action implemented.
- The optimal contract reflects the *inference principle.*

Observation 6

$f(y,e)$	y_l	y_m	y_h
e_h	0	0.5	0.5
e_l	0.5	0.1	0.4
$\frac{f(y,e_l)}{f(y,e_h)}$	∞	0.20	0.8



$t(y)$ depends on likelihood ratio. Even under FOSD optimal contract need not be monotone.

Optimal contract: $t(yl) < t(yh) < t(ym)$

Monotone optimal contract

- FOC condition for optimality is

$$\frac{1}{U_a'(t(y))} = \lambda + \mu \left[1 - \frac{f(y,e_l)}{f(y,e_h)} \right]$$

- $t(y)$ is monotone nondecreasing in y if and only if $\frac{d\left[\frac{f(y,e_l)}{f(y,e_h)}\right]}{dy} \leq 0.$

Monotone and linear contracts

- Monotone contracts require MLRP, a strong assumption.
- If agent can ``reduce” output at no cost/effort, then transfer function ought to be monotone; otherwise the agent would artificially reduce output when necessary.

In example: reduce y_h to y_m when the outcome is y_h.

- Linear contracts
- 1.
- 2.

Value of Information: Statistics preliminaries

Definition 5 Let $S \subset \mathbb{R}^n$ be a sample space. A model is a family of probability distributions $\{P_\theta : \theta \in \Theta\}$ where $(\forall \theta \in \Theta)$ P_θ is a probability distribution on S and Θ is a set, called the parameter set.

A model is regular if $(\forall \theta \in \Theta)$ either

1. P_θ is $\mathbb{C}^0$ and has density $p(x, \theta)$.
2. P_θ is discrete with frequency $p(x, \theta)$ and $\exists \{x_1, x_2, ...\} \subseteq S$ such that $\sum_{i=1}^{\infty} p(x_i, \theta) = 1$.

Note: (1) requires that every P_θ be absolutely continuous.

Statistic preliminaries

Definition 6 A statistic in the model is any function $T : S \to W$ where W is a space.

- W is often some Euclidean space or even $\mathbb{R}$.

Definition 7 A statistic T is a reduction of data if $(\exists x, x' \in S, x \neq x')$ and $T(x) = T(x')$.

- Any statistic that is a reduction of data causes a loss of information.
- A statistic T is sufficient for a parameter of interest, if observing the realization $T(x) = t$, does not imply a loss of information about the parameter of interest.

Statistic preliminaries

Definition 8 The statistics $T : S \to W$ is sufficient for θ if $(\exists Q : \mathbb{R}^n \times W \to [0,1])\ (\forall \theta \in \Theta)\ (\forall t \in W)\ [P_\theta(x|T(x) = t) = Q(x,t)]$.

- The probability of x given $T(x) = t$, does not depend on θ. Equivalently, once t is observed, knowing the sample realization x (such $T(x) = t$) does not contain additional information about θ (or P_θ) provided the model is valid.
- Determining directly whether a statistic is sufficient for a parameter is often difficult because conditional probabilities must be computed.

Statistic preliminaries

Theorem 4 (Factorization theorem) T is sufficient for θ if and only if $(\exists g : W \times \Theta \to \mathbb{R})$ $(\exists h : S \to \mathbb{R})$ such that $(\forall x \in S)$ $(\forall \theta \in \Theta)$

$$p(x, \theta) = g(T(x), \theta)h(x)$$

- The Factorization theorem holds in more general environments than the regular models presented here. For a general treatment, see Lehmann (1997).

Value of information

- The optimal contract exploits the correlation between y and e. Suppose there is another variable x that is correlated with e.
- Would a transfer $t(x, y)$ dominate the optimal $t(y)$? Yes, whenever x gives information about e that is not contained in y.

This results is due to Holmström (1979).

Value of information in principal agent model

Theorem 5 Let $t(y)$ be an optimal transfer function for which the agent's choice of action is unique and interior in A. Then, there exist $t(x, y)$ (where x is another observable variable) which strictly pareto dominates $t(y)$ if and only if x is informative about e.

Definition 9 x is informative about e whenever y is not a sufficient statistic for (x, y) with respect to e.

The result is due to Holmström (1979); the paper contains a useful discussion and the proof of the theorem.

Mirrlees and the FOA

- See Mirrlees (1999). This paper was written in 1975.
  It was broadly read as a working paper.
- See Rogerson (1985).
- See Jewitt (1988).
- See Conlon (2009).
- See Chade and Swinkels (2019).

References

Bolton, Patrick, and Dewatripont. 2005. *Contract Theory*. MIT Press.

Borch, Karl. 1962. “Equilibrium in a Reinsurance Market.” *Econometrica* 30 (3): 424–44. https://doi.org/10.2307/1909887.

Chade, Hector, and Jeroen Swinkels. 2019. “The No-Upward-Crossing Condition, Comparative Statics, and the Moral Hazard Problem.” Unpublished manuscript.

Conlon, J. 2009. “Two New Conditions Supporting the First-Order Approach to Multisignal Principal–Agent Problems.” *Econometrica* 77: 249–78.

Demski, Joel S., and David M. Kreps. 1982. “Models in Managerial Accounting.” *Journal of Accounting Research* 20: 117–48.

Diamond, Douglas W. 1984. “Financial Intermediation and Delegated Monitoring.” *The Review of Economic Studies* 51: 393–414. https://doi.org/https://doi.org/10.2307/2297430.

Grossman, Sanford, and Oliver Hart. 1983. “An Analysis of the Principal-Agent Problem.” *Econometrica* 51: 7–45.

Hart, Oliver, and Bengt Holmström. 1987. “The Theory of Contracts.” In *Advances in Economic Theory, Fifth World Congress*, edited by Truman Bewley. Cambridge University Press.

Holmström, Bengt. 1979. “Moral Hazard and Observability.” *Bell Journal of Economics* 10: 74–91.

Jensent, Michael C., and William H. Meckling. 1976. *Jounral of Finance* 3: 305–60. https://www.sciencedirect.com/science/article/pii/0304405X7690026X.

Jewitt, Ian. 1988. "Justifying the First-Order Approach to Principal-Agent Problems." *Econometrica*, 1177–90.

Lehmann, E. L. 1997. *Testing Statistical Hypotheses*. 2nd ed. Springer Texts in Statistics. Springer. https://doi.org/10.1007/978-1-4757-1923-9.

Marshall, Albert W., Ingram Olkin, and Barry C. Arnold. 2010. *Inequalities: Theory of Majorization and Its Applications*. 2nd ed. Springer. https://doi.org/10.1007/978-0-387-68276-1.

Mas-Colell, Andreu, Michael Whinston, and Jerry Green. 1995. *Microeconomic Theory*. Oxford University Press.

Milgrom, Paul R. 1981. "Good News and Bad News: Representation Theorems and Applications." *The Bell Journal of Economics* 12 (2): 380–91.

Mirrlees, James A. 1999. "The Theory of Moral Hazard and Unobservable Behaviour: Part i." *The Review of Economic Studies* 66 (1): 3–21.

Rogerson, W. 1985. "The First-Order Approach to Principal-Agents Problems." *Econometrica* 53: 1357–67.

Rothschild, Michael, and Joseph Stiglitz. 1976. "Equilibrium in Competitive Insurance Markets: An Essay on the Economics of Imperfect Information." *The Quarterly Journal of Economics* 90 (4): 629–49. http://www.jstor.org/stable/1885326.

Royden, Halsey Lawrence. 1968. *Real Analysis*. 2nd ed. McMillan Publishing Co.

Shavell, Steven. 1979. "On Moral Hazard and Insurance*." *The Quarterly Journal of Economics* 93 (4): 541–62. https://doi.org/10.2307/1884469.